User Guide

Retail & Supply Chain Control Tower

1. About the Application

The Nexvia Retail & Supply Chain Control Tower is a data-driven application designed to monitor retail supply chain operations. It helps users understand inventory availability, supplier performance, shipment delays, and product demand through dashboards and analytics.

2. How to Access the Application

1. Ensure that PostgreSQL and the required project services are running.
2. Start Apache Superset using the configured Docker environment.
3. Open the Superset application in your web browser.
4. Log in using your configured credentials.
5. Navigate to the project dashboard.

3. Using the Dashboard

Users can explore the dashboard to understand important supply chain indicators:

Dashboard Area	What Users Can Understand
Inventory Monitoring	Available stock and low-stock products
Supplier Analysis	Supplier reliability and lead-time information
Shipment Tracking	Shipment status and delivery delays
Demand Analysis	Historical product demand and trends
Forecasting	Expected future product demand

4. Understanding Key Indicators

- **Total Products:** Number of products included in inventory analysis.
- **Low Stock Products:** Products requiring attention based on their stock levels.
- **Delayed Shipments:** Shipments delivered later than their expected delivery dates.
- **Average Delay:** Average delivery delay in days.
- **Days of Inventory (DOI):** Estimated number of days current inventory can support demand.

5. Data Refresh Process

The application uses an automated hourly data pipeline.

1. Python generates updated supply chain datasets.
2. CSV files are imported into PostgreSQL.
3. The Windows Task Scheduler executes the process every hour.
4. Execution details are stored in hourly_refresh.log.
5. Updated database data can be reflected in dashboard visualizations after the dashboard data is refreshed.

Note: Database refresh and dashboard auto-refresh are separate processes. Automatic dashboard refresh depends on the Superset dashboard configuration.

6. Tips for Users

- Check low-stock products regularly to identify replenishment requirements.
- Monitor delayed shipments to identify delivery issues.
- Review supplier performance to understand reliability.
- Analyze demand trends before making inventory decisions.
- Check the latest data refresh logs when updated information is expected.

7. Troubleshooting

Issue	Suggested Action
Dashboard not opening	Check whether Docker and Superset services are running
Data not updated	Check hourly_refresh.log
Database connection issue	Verify PostgreSQL service and connection settings
Dashboard showing old data	Refresh the dashboard data and verify database updates

8. Conclusion

The Nexvia Retail & Supply Chain Control Tower provides a centralized view of key supply chain operations. Users can monitor inventory, evaluate suppliers, track shipment delays, and understand product demand to support better operational decisions.

Project Type: Simulated Retail Supply Chain Analytics
Data Storage: PostgreSQL
Dashboard Platform: Apache Superset
Automation: Python and Windows Task Scheduler